Introduction to Operations Research, Deterministic Models

Lecture 12: Nonlinear Programming

Department of Industrial Engineering, Tsinghua University

Instructor: Junyi Liu

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General Formulation of Optimization Problems

Components

- variables $\mathbf{x} \in \mathbb{R}^n$
- objective function $f(\mathbf{x})$
- constraint functions $g_i(\mathbf{x})$, $h_i(\mathbf{x})$
- constraint set $X \subseteq \mathbb{R}^n$

A general optimization problem:

Minimize	$f(\mathbf{x})$		(objective function)
subject to	$g_i(\mathbf{x}) \leq 0,$	for $i = 1, \ldots, m$	(inequality constraints)
	$h_i(\mathbf{x}) = 0,$	for $i = 1, \ldots, \ell$	(equality constraints)
	$\mathbf{x} \in X$		(set constraint)

Example 1: the basic pricing problem

Assume the demand $d(p) = 6 - 2p$ is a nonincreasing linear function of the price p for $p \in [1, 3]$. Suppose the unit production cost is 1, and the production capacity is C.

Goal: determine the production quantity q and price p, in order to maximize the profit.

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$$\begin{array}{lll} \text{maximize}_{q,p} & q \times (p-1) \\ \text{subject to} & q \leq 6 - 2p, \quad q \leq C \\ & p \in [1, 3] \end{array} \iff \begin{array}{ll} \text{maximize}_{p} & \min\{6 - 2p, C\} \times (p-1) \\ \text{subject to} & p \in [1, 3] \end{array}$$

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Extensions:

- pricing for multiple products;
- demand is uncertain, and the units to be produced need to be determined before the demand is realized (stochastic programming)
- the holding costs for the left-over units and the penalty costs for the unsatisfied demands;

Example 2: portfolio management problem

For asset i, the annual rates of return R_i are random variables with $\mathbb{E}[R_i] = \mu_i$ for $i = 1, \ldots, n$, and covariance matrix $Q = (Q_{ij})$ where $Q_{ij} = \text{Cov}(R_i, R_j)$.

So if we invest a fraction x_i of our investment dollar in asset i with $\sum_{i=1}^{n} x_i = 1$, then:

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So if we invest a fraction x_i of our investment dollar in asset i with $\sum_{i=1}^{n} x_i = 1$, then:

- **expected return** of the portfolio is $\mathbf{x}^\top \mu$;
- **standard deviation** of the portfolio is $\sqrt{\mathrm{Var}(\mathbf{x}^\top \mathbf{R})} = \sqrt{\mathbf{x}^\top Q\, \mathbf{x}}$.

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- **standard deviation** of the portfolio is $\sqrt{\text{Var}(\mathbf{x}^\top \mathbf{R})} = \sqrt{\mathbf{x}^\top Q\, \mathbf{x}}$.

Goal: maximize the expected return of the portfolio, subject to meeting some pre-specified target level of the risk.

Example 2: portfolio management problem

$$
\begin{aligned}
\text{maximize} \quad & \mathbf{x}^\top \mu \\
\text{subject to} \quad & \mathbf{x}^\top Q\, \mathbf{x} \leq \tau \\
& \sum_{i=1}^{n} x_i = 1 \\
& x_i \in [0,1], \text{ for } i = 1, \ldots, n
\end{aligned}
$$

Key insight: an asset's risk and return should not be assessed by itself, but by how it contributes to a portfolio's overall risk and return.

Example 2: portfolio management problem

Mean-Variance Portfolio Management

Economist *H*arry Markowitz introduced the Modern portfolio theory (MPT) in a 1952 essay for which he was later awarded a **Nobel Memorial Prize** in Economic Sciences in 1990.

In the preceding year, he received the **John von Neumann Theory Prize** from the Operations Research Society of America (now INFORMS).

Applications of nonlinear programming

- **revenue management:** pricing problems, advertising (online/offline) and marketing;
- **finance:** portfolio management, pricing of financial instruments, trading strategies of investment funds;
- **control:** path planning of unmanned vehicles;
- **transportation:** vehicle traffic, airline transportation;
- **machine learning:** prediction, classification, recommendation;

Nonlinear objective and constraint functions appear in most of the real-world problems

Convex sets and convex functions

Linear V.S. Nonlinear Programming

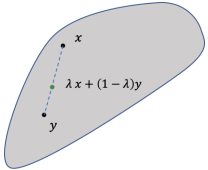
Initially, it was thought that there is a critical distinction between linear and nonlinear optimization problems.

But some nonlinear problems turned out to be much harder than others.

Now it is widely recognized that the right distinction is between **convex** and **nonconvex** problems

Convex sets

A set $S \subseteq \mathbb{R}^n$ is defined to be a convex set if for any $x, y \in S$, $\lambda x + (1-\lambda)y \in S$ for any $\lambda \in [0, 1]$.



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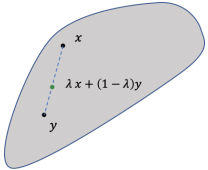


Examples:

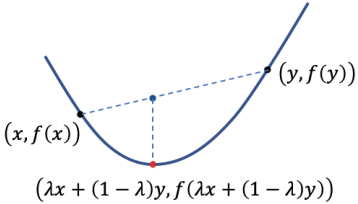
- **Hyperplane:** $S = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{p}^\top \mathbf{x} = a\}$
- **Halfspace:** $S = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{p}^\top \mathbf{x} \leq a\}$
- **Polyhedral set:** $S = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{A}\mathbf{x} \leq \mathbf{b}\} \subset \mathbb{R}^n$ where $\mathbf{A} \in \mathbb{R}^{m \times n}$ and $\mathbf{b} \in \mathbb{R}^m$
- **Intersection:** $S_1 \cap S_2$ is convex if S_1 and S_2 are convex sets;
- **Scaling and translation:** $aS + b$ is convex if S is convex;

Convex functions

Definition

Let $f : S \subseteq \mathbb{R}^n \to \mathbb{R}$ where S is a nonempty convex set. We say f is convex on S if for any $\mathbf{x}, \mathbf{y} \in S$ and any $\lambda \in (0, 1)$,

$$f(\lambda \mathbf{x} + (1 - \lambda)\mathbf{y}) \leq \lambda f(\mathbf{x}) + (1 - \lambda) f(\mathbf{y}).$$



We say f is **concave** on S if $-f$ is convex on S.

Examples

- $f_1(x) = 3x + 4$
- $f_2(x) = |x|$
- $f_3(x) = x^2 - 2x$
- $f_4(x) = x^3$

Examples

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How to verify the convexity of a function?

verification through definition is too complicated.

sufficient conditions for convexity that are easier to be verified?

Convex functions

Operations that preserve convexity

Let $f_1, f_2, \ldots, f_k : S \subseteq \mathbb{R}^n \to \mathbb{R}$ be convex functions on a convex set S. Then

- **positive combination:**

  $f(\mathbf{x}) = \sum_{j=1}^{k} \alpha_j \, f_j(\mathbf{x})$ where $\alpha_j > 0$ for $j = 1, 2, \ldots, k$ is a convex function.

- **maximum:** $f(\mathbf{x}) = \max\{f_1(\mathbf{x}), f_2(\mathbf{x}), \ldots, f_k(\mathbf{x})\}$ is a convex function.

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Twice continuously differentiable function

Let $f : \mathbb{R}^n \to \mathbb{R}$ be twice continuously differentiable.

The function f is convex $\iff$ the Hessian matrix $\mathbf{H}(\mathbf{x})$ is positive semidefinite for any $\mathbf{x} \in \mathbb{R}^n$.

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We can compute the Hessian matrix $\mathbf{H}(\mathbf{x})$ and see if

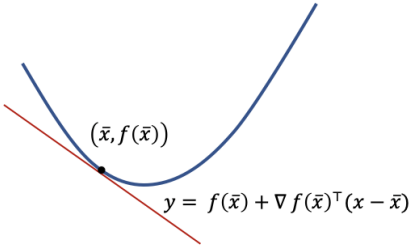
- $\mathbf{z}^\top \mathbf{H}(\mathbf{x})\mathbf{z}$ is nonnegative for any $\mathbf{z} \in \mathbb{R}^n$, and $\mathbf{x} \in \mathbb{R}^n$;
- eigenvalues of $\mathbf{H}(\mathbf{x})$ are nonnegative for any $\mathbf{x} \in \mathbb{R}^n$;

Differentiable convex functions

Theorem

Let S be a nonempty convex set in $\mathbb{R}^n$, and let $f : S \to \mathbb{R}$ be differentiable on S. Then

$$f \text{ is convex} \iff \text{for any } \bar{\mathbf{x}} \in S,\ f(\mathbf{x}) \geq f(\bar{\mathbf{x}}) + \nabla f(\bar{\mathbf{x}})^\top (\mathbf{x} - \bar{\mathbf{x}}),\ \text{for all } \mathbf{x} \in S.$$



Any differentiable convex objective function $f(\mathbf{x})$ can be lower bounded by an affine function $f(\bar{\mathbf{x}}) + \nabla f(\bar{\mathbf{x}})^\top (\mathbf{x} - \bar{\mathbf{x}})$;

Non-differentiable convex functions

Subgradient

Let S be a nonempty convex set in $\mathbb{R}^n$ and let $f : S \to \mathbb{R}$ be a convex function. For $\bar{x} \in \text{int}S$, there exists a vector ξ such that $f(\mathbf{x}) \geq f(\bar{\mathbf{x}}) + \xi^\top(\mathbf{x} - \bar{\mathbf{x}})$, for all $\mathbf{x} \in S$. The vector ξ is called a **subgradient** of f at $\bar{\mathbf{x}}$

Subdifferential

Let S be a nonempty convex set in $\mathbb{R}^n$ and let $f : S \to \mathbb{R}$ be a convex function.
The **subdifferential** of f at $\bar{\mathbf{x}}$ is the collection of subgradients of f at $\bar{\mathbf{x}}$, i.e.,

$$\partial f(\bar{\mathbf{x}}) = \{\, \xi \in \mathbb{R}^n : f(\mathbf{x}) \geq f(\bar{\mathbf{x}}) + \xi^\top(\mathbf{x} - \bar{\mathbf{x}}), \forall \mathbf{x} \in S \,\}$$

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Property

for a convex function $f : S \to \mathbb{R}$ defined on a nonempty convex set S,

- the set $\partial f(\bar{\mathbf{x}}) \neq \emptyset$ if $\bar{\mathbf{x}} \in \text{int}S$;
- If in addition f is differentiable, then $\partial f(\bar{\mathbf{x}}) = \{\nabla f(\bar{\mathbf{x}})\}$

Convex program

Convex program is a class of optimization problems which minimizes a convex function restricted to a convex set.

With $f(x)$ and $\{g_i(x)\}$ being convex functions, and X being a convex set, the following optimization is a convex program.

$$\begin{array}{lll} \text{Minimize} & f(\mathbf{x}) & \text{(objective function)} \\ \text{subject to} & g_i(\mathbf{x}) \leq 0, \quad \text{for } i = 1, \ldots, m & \text{(inequality constraints)} \\ & \mathbf{x} \in X & \text{(set constraint)} \end{array}$$

Convex constrained set

If $g_i : \mathbb{R}^n \to \mathbb{R}$ is convex for $i = 1, \ldots, m$ and $X \subseteq \mathbb{R}^n$ is a convex set, the set $\{\mathbf{x} \in X \subseteq \mathbb{R}^n : g_i(\mathbf{x}) \leq 0, i = 1, \ldots, m\}$ is a convex set.

Convex program

A general convex program:

$$
\begin{array}{lll}
\text{Minimize} & f(\mathbf{x}) & \text{(objective function)} \\
\text{subject to} & g_i(\mathbf{x}) \leq 0, \quad \text{for } i = 1, \ldots, m & \text{(inequality constraints)} \\
 & h_i(\mathbf{x}) = 0, \quad \text{for } i = 1, \ldots, \ell & \text{(equality constraints)} \\
 & \mathbf{x} \in X & \text{(set constraint)}
\end{array}
$$

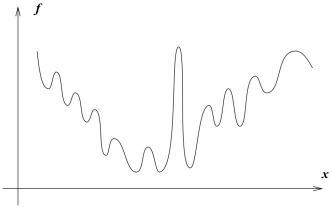
where $f(x)$ and $\{g_i(x)\}$ are convex functions, $\{h_i(x)\}$ are linear functions, and X is a convex set.

A function is convex and concave if and only if it is a linear function.

Unconstrained nonlinear programming

Unconstrained nonlinear programming

General formulation: $\underset{\mathbf{x}\in\mathbb{R}^n}{\text{minimize}}\ f(\mathbf{x})$

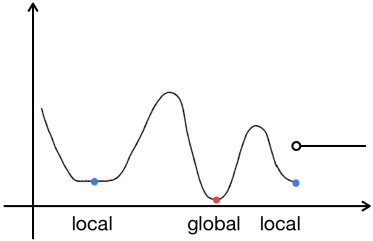


There are multiple types of optimal solutions: local optimum, global optimum, etc.

Local and global optimums

Definition

Let $f : \mathbb{R}^n \to \mathbb{R}$ be the objective function. We say $\bar{\mathbf{x}}$ is a



- **global minimizer:** if $f(\bar{\mathbf{x}}) \leq f(\mathbf{x})$ for all $\mathbf{x} \in \mathbb{R}^n$;
- **local minimizer:** $\exists\, N_\epsilon(\bar{\mathbf{x}})$ such that $f(\bar{\mathbf{x}}) \leq f(\mathbf{x})$ for any $\mathbf{x} \in N_\epsilon(\bar{\mathbf{x}})$;

- **global minimizer** $\Longrightarrow$ **local minimizer**
- **global minimizer** $\Longleftarrow$ **local minimizer** if f is a convex function

Local and global optimum

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Proof.

1. Let $\bar{\mathbf{x}}$ be a local optimum solution, then there exists a neighborhood $N_\epsilon(\bar{\mathbf{x}})$ such that $f(\mathbf{x}) \geq f(\bar{\mathbf{x}})$ for each $\mathbf{x} \in N_\epsilon(\bar{\mathbf{x}})$.
2. By contradiction, suppose that $\bar{\mathbf{x}}$ is not a global optimal solution, i.e., $\exists\, \hat{\mathbf{x}} \in \mathbb{R}^n$, $f(\hat{\mathbf{x}}) < f(\bar{\mathbf{x}})$.
3. With sufficiently small $\lambda > 0$, $\lambda\hat{\mathbf{x}} + (1-\lambda)\bar{\mathbf{x}} \in N_\epsilon(\bar{\mathbf{x}})$.
4. By the convexity of f,

$$f(\lambda\hat{\mathbf{x}} + (1-\lambda)\bar{\mathbf{x}}) \leq \lambda f(\hat{\mathbf{x}}) + (1-\lambda) f(\bar{\mathbf{x}}) < f(\bar{\mathbf{x}}),$$

which contradicts the local optimality of $\bar{\mathbf{x}}$. Hence, $\bar{\mathbf{x}}$ is a global minimizer.

Optimality conditions for unconstrained programs

Questions: given a point $\bar{\mathbf{x}}$, how to verify if it is a local minimum for a general unconstrained nonlinear program?

Necessary optimality condition

If a point $\bar{\mathbf{x}}$ is a local minimum, then xxx must hold at $\bar{\mathbf{x}}$.

Sufficient optimality condition

If xxx holds at $\bar{\mathbf{x}}$, then the point $\bar{\mathbf{x}}$ is a local minimum.

There are various optimality conditions depending on the properties of the objective function. We shall carefully select an appropriate one.

Necessary optimality conditions

For a differentiable function $f : \mathbb{R}^n \to \mathbb{R}$, $\Delta x \in \mathbb{R}^n$ denotes a small variation,

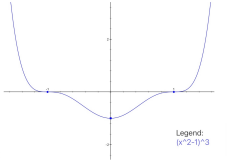
$$f(\bar{\mathbf{x}} + \Delta x) - f(\bar{\mathbf{x}}) \approx \nabla f(\bar{\mathbf{x}})^\top \Delta x$$

Theorem

Suppose that $f : \mathbb{R}^n \to \mathbb{R}$ is differentiable at $\bar{\mathbf{x}}$. If $\bar{\mathbf{x}}$ is a local minimizer, then $\nabla f(\bar{\mathbf{x}}) = 0$. (it is originally formulated by Fermat in 1637 without proof.)

The vector $\bar{\mathbf{x}}$ that satisfies $\boxed{\nabla f(\bar{\mathbf{x}}) = 0}$ is called a **stationary** solution. It could be

- a local minimizer: $f(x) = (x-1)^2$, $f'(1) = 0$
- a local maximizer: $f(x) = -(x-1)^2$, $f'(1) = 0$



Sufficient and necessary optimality condition: convex case

Suppose that f is a convex function on $\mathbb{R}^n$.

Theorem

For the unconstrained program minimize$_{\mathbf{x}\in\mathbb{R}^n}$ $f(\mathbf{x})$

$$\bar{\mathbf{x}} \text{ is a global(local) minimizer} \iff 0 \in \partial f(\bar{\mathbf{x}}).$$

If f is additionally differentiable,

$$\bar{\mathbf{x}} \text{ is a global(local) minimizer} \iff \nabla f(\bar{\mathbf{x}}) = 0.$$

Why do we need optimality conditions?

- use optimality conditions to find solutions of an optimization problem

1. find all points that satisfy $\nabla f(x) = 0$
2. select the one with the minimal cost

- however, solving the system of equation $\nabla f(x) = 0$ is nontrivial. Algorithmically, it is typically as difficult as solving the original optimization problem

- optimality conditions provide the basis for the development and analysis of algorithms

Algorithms for nonlinear programming

Algorithm:
an iterative process that generates a sequence of points according to a prescribed set of instructions, together with a termination criterion.

Strategies:
• most algorithms make use of the values of the objective function and constraint functions, and possibly first-order and second-order derivatives and subgradients of these functions.

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Strategies:
• most algorithms make use of the values of the objective function and constraint functions, and possibly first-order and second-order derivatives and subgradients of these functions.

• some algorithms use only local information obtained at the current iteration; while others accumulate information gathered at previous iterations to strengthen the convergence

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- in practice, as a result of nonconvexity, problem size, randomness, and other computational difficulties, we may have to be satisfied with some solution sets that are weaker than global optimality (e.g., some solutions that satisfy the necessary optimality conditions)
- recognize and analyze the optimality conditions that the solution sequence provided by algorithm, and terminate the algorithm when optimality conditions hold approximately

Iterative descent methods for nonlinear programming

1. **Initialization:** start at some initial point $\mathbf{x}^0$;

2. **Iterative process:** iteratively generate a sequence of vectors $\mathbf{x}^1, \mathbf{x}^2, \ldots,$ according to

$$\boxed{\mathbf{x}^{r+1} \leftarrow \mathbf{x}^r + \alpha^r \mathbf{d}^r}$$

- step size: α^r
- direction: $\mathbf{d}^r$
- iterative solution: $\mathbf{x}^r$

3. **Descent property:** the step sizes $\{\alpha^r\}$ and directions $\{\mathbf{d}^r\}$ are chosen such that f is decreased at each iteration, i.e.,

$$f(\mathbf{x}^{r+1}) = f(\mathbf{x}^r + \alpha^r \mathbf{d}^r) < f(\mathbf{x}^r)$$

hope to decrease f all the way to the stationary solutions or (local) minimum.

Reading materials

Section 7.1, 7.2, 7.5, *Operations Research: mathematical programming* by Wayne L. Winston